

Synergizing hybrid renewable energy systems and sustainable agriculture for rural development in Nigeria

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ABSTRACT

This research explores the viability and importance of implementing Hybrid Renewable Energy Systems (HRES) in Nigeria's agriculture sector. Despite progress, most rural regions still struggle with unreliable electricity. HRES provides a way to power agricultural endeavors while promoting sustainability, economic viability, and ecological conservation. A case study involving renewable energy applications for powering a typical Nigerian farm facility is also presented. Based on the review's findings, while 53.1 % of the journals analyzed were primarily concerned with community load, the agricultural sector received much less coverage, especially in rural areas. For the case study, solar PV dominates the HRES (98.3 %) electricity production, with biogas and wind turbines contributing the rest. The project costs \$1,782,602 over 25 years, with competitive energy costs and a discounted payback time of 7.22 years. Environmentally, carbon dioxide emissions are negligible (16 kg/yr), and excess electricity production offers income potential for the farm facility. Overall, HRES is presented as an environmentally beneficial and cost-effective way to change rural farm operations, emphasizing the critical need to integrate energy and agriculture. Still, challenges such as inaccurate data, inadequate infrastructure, and inconsistent policy dynamics remain. In addition, the work highlights an area where more study is needed, namely the use of MCDM analysis for selecting the optimal energy system based on multiple criteria. Utilizing HRES in agriculture is consistent with the SDGs and will increase agricultural productivity, self-sufficiency, and environmental friendliness in Nigeria.

1. Introduction

Nigeria, a nation renowned for its extensive cultural heterogeneity and bountiful endowment of natural resources [1], simultaneously contends with intricate predicaments that profoundly impact the day-to-day existence and welfare of its rural communities. Two significant concerns on the national agenda are the persistent shortage (lack in some cases) of electricity in rural regions and the necessity for sustainable agriculture methods [2]. The interconnection of these challenges

necessitates their effective resolution to foster the country's development, stimulate economic growth, and ensure environmental sustainability.

Nigeria's large, diverse terrain is home to a substantial rural population for whom access to affordable and reliable energy services remains a persistent concern. Rural settlements often experience a lack of connectivity to the national electricity grid, if such a link exists at all [3]. The perennial lack of electricity and modern energy services in these locations profoundly impacts the overall well-being of the population.

Abbreviations: HRES, Hybrid Renewable Energy Systems; SDG, Sustainable Development Goal; MTNDP, Medium-Term National Development Plan; REP, Rural Electrification Policy; RESIP, Rural Electrification Strategy and Implementation Plan; ETP, Energy Transition Plan; ERGP, Economic Recovery and Growth Plan; ESP, Economic Sustainability Plan; REA, Rural Electrification Agency; MW, Mega Watt; kW, kilo Watt; kWh, kilo Watt Hour; SSL, Solar Street Lights; SHS, Solar Home Systems; DC, Direct Current; AC, Alternating Current; FC, Fuel Cell; PV, Photovoltaic; HOMER, Hybrid Optimization Model for Electric Renewable; MATLAB, Matrix Laboratory; IoT, Internet of Things; AI, Artificial intelligence; NASRDA, National Space Research and Development Agency; COE, Cost of Energy; TNPC, Total Net Present Cost; NGO, Non-Governmental Organization; MCDM, Multicriteria Decision Making.

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Insufficient lighting conditions contribute to restricted educational access, hindered healthcare provision due to unpredictable power supply, and challenges rural industries face in achieving sustainable growth [4]. An energy deficit exacerbates a cycle of poverty, impeding the capacity of individuals residing in rural areas to break free from conditions of underdevelopment [5].

Meanwhile, Nigeria’s agricultural industry, a cornerstone of the country’s economy, faces numerous challenges. Primary barriers to agricultural output include obsolete farming methods, significant post-harvest losses, insufficient irrigation infrastructure, and the ever-looming threat of climate change [6]. Furthermore, agricultural enterprises are more susceptible since they use traditional energy sources like diesel generators. The increased expenses of operations, the decreased viability of farming techniques, and the increased environmental burdens all result from this reliance.

Considering these pertinent problems in rural energy and agriculture, developing Hybrid Renewable Energy Systems (HRES) is crucial [7]. HRES is a game-changer because of the myriad opportunities renewable energy sources incorporate [8]. These include solar, wind, hydro, biomass, advanced energy storage, and grid control technologies. These systems can harness Nigeria’s plethora of renewable resources, improve the country’s energy diversification, and lessen the adverse effects of traditional power plants on the environment [9].

However, the importance of hybrid renewable energy systems extends beyond the mere generation of energy [10]. Implementing these solutions holds the capacity to revolutionize the agricultural terrain in rural Nigeria. HRES can enable precision farming, support efficient irrigation practices, and facilitate value-added processes such as food preservation and agro-processing by providing a reliable and sustainable energy source [11]. This aligns the HRES with sustainable agriculture principles, in which ecological responsibility and economic viability coexist. In addition to increased yields and prudent resource utilization, sustainable agriculture can improve rural livelihoods, stimulate economic development, and promote social equity [12].

This survey aims to examine the complex relationship between Hybrid Renewable Energy Systems (HRES) and sustainable agriculture in Nigeria in the context of rural development. It aims to provide a nuanced comprehension of how HRES can effectively power agricultural activities sustainably, promoting rural development and economic prosperity. In addition, this review seeks to identify the obstacles, opportunities, and innovations associated with incorporating HRES into agricultural applications. It will analyze in minute detail the multifaceted effects of such integration on rural livelihoods, agricultural productivity, and environmental sustainability.

The current study provides various innovative advances compared to prior research on agricultural electrification and hybrid renewable energy systems (HRES). This paper extensively discusses several topics, including opportunities, challenges, technological advancements, agricultural applications, economic and environmental impacts, case studies, and relevant policy and regulation discussions. The exclusive emphasis on the economic and environmental impacts of HRES application in the agricultural sector is unique, with only a few previous

papers exploring both aspects simultaneously. Furthermore, the paper presents a case study on the application of HRES for powering an existing farm facility. A comparison of the present study to other similar studies in the literature is presented in Table 1. The paper also contributes to the interconnections between the Sustainable Development Goals (SDGs) related to agriculture, energy, and other SDGs. The matrix in Fig. 1 illustrates the interconnections between the SDGs on agriculture (Goals 2 and 15) and energy (Goals 7 and 13). For instance, there is a clear correlation between sustainable agriculture and the eradication of hunger and the enhancement of agricultural output through energy access. Increasing the use of clean energy contributes to promoting sustainable production. Furthermore, there are significant links to climate change mitigation concerning these subjects.

The scope of this review includes examining HRES technologies, their application in rural Nigeria, and the complex economic, social, and environmental implications of these systems within the agricultural sector. Furthermore, it will examine the existing policy and regulatory frameworks that either facilitate or hinder HRES adoption in agriculture. This review aims to contribute substantially to the discussion on sustainable rural development in Nigeria by synthesizing existing knowledge, casting light on critical issues, and identifying gaps in research and practice. The primary objective of this study is to motivate future research undertakings, policy initiatives, and practical interventions that effectively tackle these significant concerns. Doing so will contribute to the advancement of rural communities in Nigeria, leading to a more promising and sustainable future.

2. Nigeria’s energy landscape and rural electrification

Energy problems, especially in the country’s rural parts, persist despite Nigeria’s vast oil and gas reserves. Even though Nigeria has the biggest economy in Africa, many people still do not have access to electricity. Most of this highly energy-deficient population lives in the rural areas [20]. Though connected to the electrical grid, the cities do not get adequate access to electricity. This challenge has encouraged more rural-urban migration for a better life by its citizens. A significant population of Nigerians still live in rural areas with a very high tendency of not having access to electricity [21]. Because of this challenge, rural areas lag in economic growth, agriculture, health, and education [22].

2.1. Rural electrification access rates and challenges

According to [23], over 90 million Nigerians reside in rural communities, primarily unconnected to the national grid or mini-grid solutions [24]. With an average generation capacity of 4,000MW serving a population of 200 million individuals, a significant portion of the Nigerian population is deprived of access to electricity. Furthermore, those who do have access often have restricted hours of availability. While the level of grid connectivity is relatively equitable in urban areas, rural regions have more significant challenges in this regard. Based on the World Bank’s compilation of development indicators, it is evident that a considerable majority, precisely 66.9 percent, of individuals

Table 1
Comparison of previous studies and the present study.

Ref	Rural Electrification	HRES Discussion	Opportunities & Challenges	Advancement in HRES & Agricultural Application	Economic & Environmental Impact	Case Study	Policies & Discussion
[13]		✓			✓		✓
[14]	✓	✓					
[15]		✓		✓			
[16]		✓	✓		✓		
[17]	✓	✓			✓		✓
[18]		✓					✓
[19]	✓	✓		✓			
Present Study	✓	✓	✓	✓	✓	✓	✓

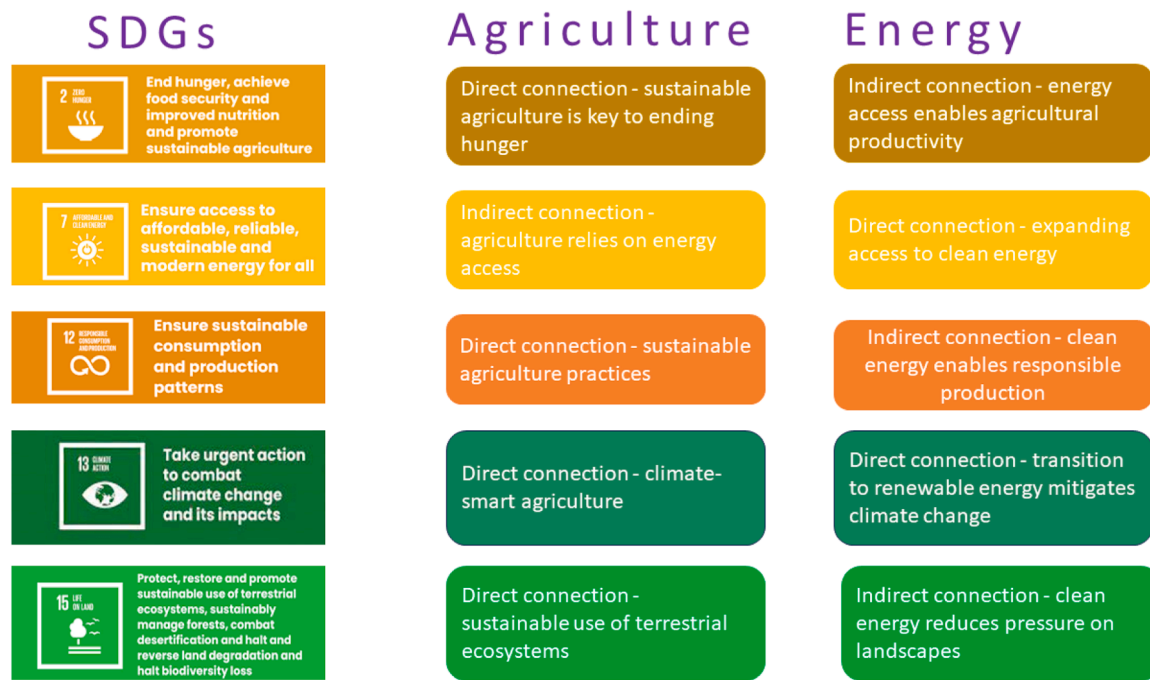


Fig. 1. Matrix showing the interrelationship between the SDGs related to agricultural (Goals 2 and 15) and energy (Goals 7 and 13).

residing in rural areas in Nigeria lack access to electricity as of the year 2020 [25]. Of this figure, a significant proportion of rural households in Nigeria continue to rely on traditional energy sources for lighting and cooking, including biomass (wood, crop residues) and kerosene. In addition to posing health risks due to domestic air pollution, these energy sources are also environmentally unsustainable. In rural areas, the shortage of access to electricity hinders the development of small and medium-sized businesses, thereby impeding economic growth and job creation.

In 2013, the Nigerian government implemented the privatization of the electricity sector. Concurrently, the Federal Government of Nigeria formed the Rural Electrification Agency (REA) with the intention of addressing the absence of power in rural communities. These areas suffer from inadequate access to electricity, which hampers several essential activities such as residential living, agricultural practices, healthcare services, and socio-economic endeavors.

Despite the existence of various governmental initiatives such as the Medium-Term National Development Plan (MTNDP), Rural Electrification Policy (REP), Rural Electrification Strategy and Implementation Plan (RESIP), Energy Transition Plan (ETP), Economic Recovery and Growth Plan (ERGP), and Economic Sustainability Plan (ESP), the primary focus has been on the provision of electricity to achieve the objectives outlined in these plans. Furthermore, it is imperative to integrate the goals of rural electrification into economic and development strategies. Based on these initiatives, during the period spanning from 2020 to 2022, Nigeria allocated a total expenditure of approximately N45.89 billion towards the provision of power infrastructure for rural communities [26]. Based on reports disseminated by the Rural Electrification Agency (REA), the initiative has executed more than 2000 electrification endeavors, influencing the livelihoods of approximately 5000,000 individuals in Nigeria. Because of these capital projects, the cumulative installed capacity is estimated to exceed 600 MW, equivalent to facilitating over 1000,000 connections.

2.2. Causes of low access rate of rural electrification

As mentioned in the previous sub-section, the rural electrification rates in Nigeria are characterized by a persistently low level, wherein a

significant number of rural families heavily depend on traditional energy sources like biomass (wood, crop leftovers), as well as kerosene for their lighting and cooking needs. These energy sources present health risks because of indoor air pollution and exhibit environmental unsustainability. The absence of electricity access poses a significant obstacle to the advancement of small and medium-sized firms in rural regions, impeding both economic growth and the generation of employment opportunities.

Several challenges contribute to the relatively low rates of rural electrification in Nigeria; these include:

- **Infrastructure limitations:** Expanding the central grid to geographically isolated rural communities poses significant financial and logistical obstacles. Consequently, numerous rural areas lack access to electricity infrastructure.
- **Economic Constraint:** The high prices associated with electricity pose a significant barrier to access for rural households, even in regions where grid connections are accessible. This financial burden renders power unaffordable for a substantial portion of the population. Economic restrictions are a significant barrier to using renewable energy in Nigeria. The high initial investment cost, coupled with her citizens' varying purchasing powers, leaves electricity sources through renewable energy beyond the reach of so many of her citizens.
- **Maintenance and reliability:** The current infrastructure for rural electrification is frequently subject to inadequate maintenance practices, resulting in frequent power outages and a diminished level of trust in the dependability of electricity provision.
- **Insufficient funding:** The availability of cost-effective financing alternatives for rural electrification initiatives is limited, posing challenges for local communities and entrepreneurs seeking to invest in renewable energy solutions.
- **Lack of Political Will:** A fundamental obstacle to Nigeria's adoption of renewable energy is a lack of political will. According to Couture and Leidreiter [27], the lack of political will is stopping the world from implementing 100 % renewable energy. Other challenges could be there but are minor compared to the lack of political will. This is evident in the Nigerian system, which lacks the proper regulatory

framework, legislation and standards that will enhance the deviation of her primary energy source from the usual conventional to renewable energy sources.

- **Social Barrier:** Social barriers have a way of posing as a restriction to development. This could be worse in rural areas. It sometimes limits positive change to development. This is because people will always resist and oppose changes that are different from their conventional way of living. This is experienced due to low knowledge in the subject area, cultural and religious beliefs, lack of awareness, etc. This unwillingness to adapt to change stands as one of the obstacles preventing Nigeria from adopting renewable energy.
- **Data Documentation:** Independent renewable energy facilities in Nigeria are not well documented. This has made it difficult to give the power rating of the number of standalone renewable energy installations operational nationwide. However, more citizens are utilizing renewable energy technology to generate electricity to mitigate the effects of unreliable electricity in their households and businesses [28]. Reliable and accurate renewable energy data that is accessible will be highly instrumental for sound policymaking on renewable energy [29].
- **Vandalization and Theft:** Vandalization and outright theft are significant problems confronting hybrid renewable energy system installations in Nigeria, though few cases have been reported compared to other African countries [30]. Civil unrest in this area can hinder the adoption of this technology in solving the country's electricity problem. Criminally-minded individuals take advantage of the breakdown of law and order to steal and vandalize HRES installations. This issue is essential since infrastructural development is only safe in a secure and stable society.

2.3. The role of renewable energy in addressing energy access in rural Nigeria

Renewable energy sources, such as solar, wind, hydro, and biomass, provide significant potential in mitigating the energy accessibility problems encountered in rural areas of Nigeria. As mentioned earlier, the sources are abundant and can be utilized locally, hence minimizing reliance on centralized power generation and fossil fuels. Renewable energy technologies, specifically off-grid and mini-grid systems, are viable and adaptable alternatives for the electrification of rural areas [31].

Solar power has experienced significant growth in rural Nigeria, with many efforts and projects focused on delivering localized solar energy solutions to areas that lack access to electricity and are neglected [9]. Solar home systems, microgrids, and solar-powered water pumping systems exemplify pioneering applications that possess the capacity to revolutionize rural areas through the provision of clean, dependable, and sustainable energy access [32]. In addition, solar solutions can be complemented by using wind and hydro energy sources. Incorporating these renewables into hybrid energy systems can improve the reliability of rural electrification efforts by providing a more steady and continuous power supply [33].

According to reports [34], in 2022, the Projects Department within the Technical Services Directorate of the Rural Electrification Agency (REA) purportedly implemented a pragmatic approach that provided electricity to six settlements. This was achieved by the installation of a 100-kW solar mini-grid system. The effort significantly impacted approximately 8155 individuals and 5000 currently engaged in farming activities, providing them with access to electricity and clean, affordable water. This resulted in various positive outcomes, including creating direct and indirect employment opportunities, enhanced security, higher productivity, improved healthcare services, and decommissioning of more than 40 diesel and petrol generators. As indicated by both present and projected estimates, the decreases in carbon emissions are also promising.

In addition, 1392 solar-powered irrigation pumps were allocated

among the six geopolitical zones, benefiting 1300 male and 92 female individuals, together with around 200 agricultural clusters. The REA revealed that the distributed facilities directly impacted the lives of more than 11,000 individuals, including over 6000 farmers, with around 810 female farmers. The aforementioned impact has resulted in the provision of Solar Street Lights (SSLs) to illuminate more than 170 farms, training to over 3000 farmers on pump maintenance and innovative irrigation practices, and significant cost reductions in maintenance, fuel, and related expenses. In the latter context, it is anticipated that the substitution of diesel-powered pumps with solar-powered pumps will increase average revenues ranging from N150,000 to N300,000.

The same year, 2046 Solar Home Systems (SHS) were implemented across the country's six geopolitical zones. These systems successfully provided access to electricity for 1972 households, as well as five primary healthcare centers, ten educational institutions, 22 religious establishments, and seven office spaces. According to the report, uninterrupted electricity directly influenced more than 10,000 lives and 5000 students. This resulted in several positive outcomes, including creating over 350 direct and indirect jobs, improved healthcare services, and enhanced trading and domestic activities among women.

Overall, renewable energy is crucial in mitigating the energy access disparity in rural regions of Nigeria [35]. By strategically utilizing Nigeria's enormous renewable resources and implementing sustainable rural electrification initiatives, the nation can enhance the standard of living within its rural communities, foster economic progress, and make valuable contributions to environmental sustainability [36].

3. Hybrid renewable energy systems in rural Nigeria

3.1. Hybrid renewable energy systems

Combining two or more renewable or non-renewable energy sources is a hybrid renewable energy system. Depending on the design, planning, and need, this might contain energy storage systems, power converters, and other electronic power devices. Its configuration complements the various energy subsystems to improve the hybrid system's consistency, efficiency, and operations. Depending on the energy combination, several hybrid energy system arrangements can be applied for power generation. The hybrid system's possible arrangements are hybridized, DC-coupled, or AC-coupled [37]. The required arrangement will be selected in line with the application. A DC-coupled system, for example, will consist of DC sources and load. In this kind of system, there will not be any need for system synchronization, and there is a lack of power quality problems, such as reactive power and harmonics [38,39]. Additionally, the system technique applies to DC microgrid.

In the connected AC system, through a variety of power electronic devices and converters, the sources are linked to a single AC bus bar [40–43]. There is a need for system synchronization in this configuration, and issues of power quality, such as harmonics and reactive power, are this configuration's main issues [44,45]. The AC-coupled system is divided into the power frequency AC coupled system and the high-frequency AC coupled system. The power frequency AC coupled system uses a variety of AC sources and loads, which improves system protection. In contrast, the high-frequency AC coupled system uses AC sources that operate at various frequencies along with high-frequency loads, improving system protection.

The connected hybrid system combines the advantages of the DC and AC coupled systems together with its characteristics of flexibility and cost-effectiveness to cumulative loads and their various sources, which depend on their characteristics [39,46,47]. In the hybrid connected system, the sources and loads can be either AC or DC connected directly or with a simple conversion stage [48,44]. This will improve flexibility, reliability, efficiency, emissions, economics, and operations compared to DC-coupled AC systems [49,50]. A hybrid technology solution is a

complex application and has to be carefully assessed. In most cases, due to the deficiency in the optimal design or proper planning and sizing of a hybrid system, it is oversized or poorly planned and designed, increasing the installation cost. A sound technical and economic assessment of a hybrid system is crucial for efficiently using the various alternative energy sources [51]. Fig. 2 gives a diagrammatic description of a hybrid renewable energy system and its essential components and subsystems.

3.2. Energy storage systems

The function of energy storage in hybrid systems and sustainable off-grid power structures cannot be neglected. Energy storage systems are devices used to stock energy for after-use. The resources of most renewable power generation through HRES primarily depend on nature and vary from time to time. This means that at every particular point in time, the energy produced will likely not be equal to the energy demand of the consumer, as it will either be higher or lower. When the hybrid system's energy supply is high without a storage system, it leads to continuous wastage of energy if there is no storage system to store the unused energy.

Conversely, if the energy supplied is lower than the energy demand of the consumer, it will result in energy deficiency, reduced reliability, and efficiency at that particular point in time. Energy storage systems are utilized to address this issue and always supply the desired load. The surplus energy created by the system during times of high or low energy demand is transferred to energy storage systems as stored energy and supplied to the load demand when the hybrid system produces less or no energy. The energy storage system is utilized to solve the problem of intermittency and discrepancies in the production and consumption of energy for a more reliable and efficient hybrid system.

In energy storage systems, the technology can either be electrical or thermal [52]. In electrical energy storage systems, the input and output are electrical quantities, while in a thermal energy storage system, the input and output are thermal quantities. Devices for storing electrical energy can be divided into electrochemical systems, for example, batteries, kinetic energy storage systems (flywheel or potential energy storage systems), and pumped hydro. High power density, superconducting magnetic energy storage (SMES), and ultra-capacitors can also play a similar role in electrical energy storage for short rapid response requirements. The thermal energy storage system stores excess heat after use under different conditions such as temperature [53]. It has both environmental and economic advantages due to the reduced need

to burn fuel. Several recent studies have shown that hydrogen storage systems composed of electrolyzed-large scale hydrogen tank-fuel cell (FC) combinations have been given attention as a promising technology.

3.3. Practical applications of renewable energy in Nigeria

There has been increased consciousness globally as it regards renewable energy and energy efficiency, which is very important in not only curtailing the issue of climatic change but also for fostering fresh, advantageous economic developments and provision of clean and affordable energy to billions of people who are still living in the absence of modern energy services [29]. When considering the limitations of the rise in the price of conventional and traditional forms of energy, renewable energy has been seen as a potential solution to Nigeria's energy poverty and deficit for people who live primarily in the country's rural areas where electricity is regarded as a luxury [54]. Nigeria is endowed with a wealth of natural resources that, when correctly utilized, may generate power. One feature of these natural resources is the renewable energy that produces clean, affordable, effective, and environmentally friendly electricity for daily necessities and economic activities. Energy is one of the measurement indexes for a prospering nation and dramatically assists economic development. This would be far greater if renewable energy sources were effectively utilized for electricity production, especially for people living in remote locations with little or no connectivity to the national grid. Rural communities' lack of access to the national grid has been a problem for a while and does not look like it will end soon. Nigeria is a country of enormous renewable energy resources such as solar, wind, biomass, and hydro-power, with varying abilities of these resources at different locations of the country; these resources, when used optimally, can be used to solve the country's energy problem, especially the rural areas that are the worst hit. Renewable energy in Nigeria has not been optimally explored in solving her electricity problem. If this is allowed, there will be tremendous improvement across all economic sectors, not just the electrical sector. This will boost agriculture, reestablish major industries, improve technology, and reinforce the nation's socioeconomic stability [55]. To accomplish the challenge of providing reliable and affordable electricity to its citizens, micro grids of renewable energy systems have to be adopted and implemented optimally to achieve this remarkable feat.

According to Chiemeka and Chineke [56], the power generated by the sun in one day is equivalent to 5.08×10^{12} kWh, and this amounts to

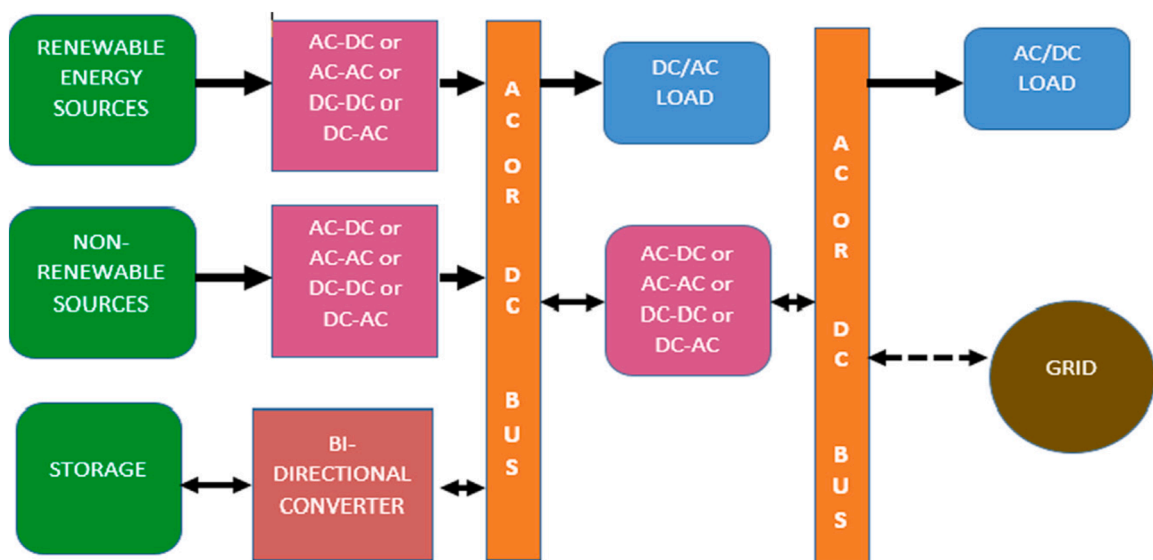


Fig. 2. Basic components of a HRES [37].

4.66 million barrels of oil per day. In line with the goals of the SDG of sustainable energy, Nigeria has not been sidelined in the campaign for the adoption of renewable energy and its energy efficiency and savings programs, even though her primary focus is to increase her electricity-generating capacity. The main features of renewable energy and its efficiency are its reduction in environmental effects, reduced cost of energy consumption, decreased emission of greenhouse gases, and accessibility. In Nigeria, various forms of renewable energy are used to power homes, streetlights, hospitals, schools, and other public places.

Some practical case studies that emphasize the efficacy of renewable energy deployment for agricultural and rural applications are presented in Table 2. These case studies demonstrate the viability of HRES in solving the energy problems plaguing Nigeria's rural areas. They demonstrate the wide range of strategies, technologies, and energy-generating potential of HRES projects in bringing prosperity to rural areas, boosting agricultural output, and aiding in sustainable development. These efforts help close the energy gap and give rural communities the tools they need to thrive economically and protect the environment.

3.4. Comparison of studies on renewable energy adoption in Nigeria

Table 3 presents a comprehensive synthesis of research conducted on hybrid renewable energy systems in Nigeria, shedding light on the diverse energy requirements of the country. These demands encompass a wide range of sectors, including remote healthcare, agricultural applications, and power solutions for universities and communities. The studies emphasize the possibilities of hybrid topologies integrating renewable sources, such as photovoltaic (PV), wind, hydro, and biomass, with conventional diesel generators. These studies utilize modeling software such as HOMER and MATLAB to achieve design and optimization. An analysis of the table shows that many studies employ system modeling and optimization software like HOMER and MATLAB. These resources make evaluating multiple potential outcomes and setups easier, making the results more solid overall.

In addition, the incorporation of sensitivity assessments, which consider variables such as diesel prices, sun radiation, and interest rates, highlights the flexibility and robustness of the suggested systems. Significantly, the concept of economic feasibility is a recurring subject matter, as evidenced by several studies assessing a range of financial indicators to ensure that planned investments align with prevailing economic conditions. The studies also address environmental factors, explicitly highlighting the reduced carbon emissions of renewable energy sources. Furthermore, the emphasis on rural and community settings caters for the urgent energy need of neglected regions and implies a commitment to achieving broader electrification and sustainability objectives.

Fig. 3 gives the percentage representation of HRES for different purposes in Nigeria for the journals reviewed. The result showed that 53.1 % worked on community load, 15.2 % on residential, 13.6 % on agricultural, 9.1 % on academic, 4.5 % on health, and 1.5 % on BTS, systems, and unspecified loads, respectively. From the papers considered, agriculture, especially in rural areas, has not been given the necessary attention it deserves to solve its electricity problem. This is because if the electricity problem of agricultural facilities is solved in the rural areas of Nigeria, it will have a ripple effect on economic empowerment, poverty alleviation, and improved standard of living, discouraging rural-urban migration, unemployment, and underdevelopment.

At present, due to the unavailability of the grid network in many rural areas, the high level of infrastructural deficit, the reluctance of investors to invest in these areas due to its low-level industrialization, and the enormous time it will take to break even in investments, the vast amount of time and finance it will consume for the provision of these infrastructures and the bureaucracy involved in its approval, the HRES remains the most viable, cost-effective and environmentally friendly solution in Nigeria [130] for rural farm facilities.

Table 2

Practical case studies renewable energy application for rural and agricultural application is Nigeria.

Ref	HRES project	Location	Description	Energy Generation	Impact
[57]	Solar Mini-Grids in Bayero University	Kano state	- Installed solar panels with battery storage to provide electricity to the university	7.1 MW	It has significantly improved living standards. - Enabled extended study hours for students.
[58]	Community-Managed Solar Microgrid in Abuja	Abuja	- Established a solar microgrid managed by the community to provide electricity for lighting, refrigeration, and small-scale businesses.	5.6 MW	- Empowered the local community. - Improved access to essential services. - Fostering economic growth and development.
[59]	Solar Power Plant and Clean Water Supply Facilities	Osun state	- Utilized solar energy to provide electricity and water pumping for the community.	50 kW	- Empowered the local community. - Improved access to essential services. - Fostering economic growth and development.
[60]	Solar Hybrid Mini-grid in Oloibiri	Bayelsa State	- Utilized solar energy for the provision of electricity for the community	67.32 kW	- Significantly improved living standards. - Empowered the local community.
[61]	Solar Hybrid Mini-grid in Akipelai	Bayelsa state	- Utilized solar energy for the provision of electricity to the community	67.32 kW	- Significantly improved living standards. - Empowered the local community.
[62]	Solar Hybrid Mini-grid in Rokota	Niger state	- Utilized solar energy for the provision of electricity to the community	64 kW	- Significantly improved living standards. - Empowered the local community.
[63]	Solar Hybrid Mini-grid in Shimankar	Plateau state	- Utilized solar energy for the provision of electricity to the community	234 kW	- Improved access to essential services. - Fostering economic growth and development.

Nevertheless, utilizing local data sources in modeling can enhance the realistic nature of the analysis. However, it is essential to acknowledge that the availability and accuracy of such data sources may present certain obstacles. Although the studies do not provide an in-depth analysis of policy or regulatory dynamics, it is widely acknowledged

Table 3

Comparison of the themes and outcomes of the articles reviewed in this study.

Ref	Type of load	Location	Aim	Energy sources considered	Hybrid	Energy storage considered	Method/Software	Uncertainty considered/ sensitivity analysis	Findings
[64]	Community	Urban	Designing an optimum off-grid energy system for isolated areas.	PV/Wind/Diesel	Yes	Battery	HOMER	–	Solar and wind energy are developable.
[65]	Healthcare	Rural	Sizing of a system of energy that is off-grid for a remote healthcare center	PV, wind turbine, diesel engine	Yes	Battery	HOMER	Interest rates, load demand, diesel price	Energy cost is directly proportional to load increase and increase in diesel price.
[66]	Community	Urban/Rural	Analyzing a standalone system of hybrid energy for the techno-economic study of the Nigerian telecom sector	PV/Diesel	Yes	Battery	HOMER	Diesel price, solar radiation	The findings showed that a properly sized PV, battery, and diesel generator would lower the net present cost and energy cost.
[67]	Residential	Rural	A PV/Hydrogen energy system was modeled and simulated for an off-grid load	PV/Hydrogen	Yes	Battery	SOHYSIMO	Solar radiation, hydrogen	The energy generated powered a cooker to augment the required estimated daily for one year.
[68]	Community	Rural	Madangho-Escravos was the research site, and the hybrid system was optimized using power energy resources.	PV/Wind/Diesel	Yes	Battery	HOMER	Price of diesel, solar radiation, wind speed.	When the study is implemented, it will be inexpensive and effective at supplying electricity to isolated rural communities and large cities.
[69]	Community	Rural	Developing a PV-micro hydro off-grid power-producing system and determining its viability	PV/Hydro	Yes	Battery	HOMER	Solar radiation, energy output, flow rate, temperature.	The selected hybrid is a feasible off-grid solution to the community's electricity problems.
[70]	Academic	Urban	The selected hybrid is a feasible off-grid solution to the community's electricity problems.	PV/Diesel	Yes	Battery	HOMER	Diesel price, solar radiation, carbon dioxide emission, interest rate	The study indicated that HRES is a better option than a diesel generator.
[71]	–	Urban/Rural	Enhancing renewable energy by fusing parts through the various energy storage technologies.	PV	No	Battery	Manually	–	The study indicated that proper device operation would provide regular electrical energy to the load.
[72]	Community	Rural	Making a hybrid system out of biomass and PV and its abilities for Kaduna, Nigeria.	PV/Biomass	Yes	Battery	Mathematical modeling	Biomass output	The study revealed that the hybrid micro grid could be reliable and have the ability to supply electricity regularly to the community.
[73]	Residential	Rural	Analyzing the parts and optimizing the hybrid system for different zones in Nigeria.	PV/Wind/Biomass	Yes	Battery	HOMER, MDCM analysis, TOPSIS-AHP	Discount rate	The study revealed that it will be affordable and favorable to rural households.
[74]	Residential	Rural	Application of PV/Hydro/Diesel generator for powering rural areas.	PV/Hydro/Diesel	Yes	Battery	HOMER	Solar radiation, diesel quantity, stream flow rate	Hydro has a significant ability to meet the village's present and near-future energy demand
[75]	Community	Rural	Analyzing the NERC's tariff through its techno-economic study for supplying power to different locations in Nigeria's northern region.	PV/Wind	Yes	Battery	HOMER	Interest rate	According to the report, the right economic incentives can make RE for grid generation profitable in Nigeria.
[76]	Residential	Urban	The abilities of PV systems for houses in Lagos, Nigeria	PV	No	Battery	HOMER	Discount rate, minimum battery SOC, inflation rate, system lifetime,	The study revealed that the number of PV panels will be reduced

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Table 3 (continued)

Ref	Type of load	Location	Aim	Energy sources considered	Hybrid	Energy storage considered	Method/Software	Uncertainty considered/ sensitivity analysis	Findings
[77]	Community	Rural	Comparing and analyzing the use of biogas and PV	PV/Biogas	Yes	–	Matlab	maximum capacity shortage –	if efficient loads are applied. The result revealed that, compared to solar energy, producing energy from biogas is more efficient and dependable for the community.
[78]	Community	Rural	The possibility of the hybrid system and its techno-economic study on rural Nigeria.	PV/Wind/ Diesel	Yes	Battery	HOMER	Solar PV radiation, speed of wind, and discount rate	The outcome demonstrated a connection between NPC and COE. There was a reduction in greenhouse gases.
[79]	Community	Rural	Analyzing a PV-powered plant for a chimney in rural Nigeria	PV	No	–	Matrix mathematical modeling	Solar PV radiation, wind speed, output energy, and diesel price.	The study revealed that access to energy can be promoted through SCPP.
[80]	Community	Rural	Analyzing the economics of the hybrid system for rural Nigeria's various zones.	PV/Wind/ Diesel	Yes	Battery	HOMER	Diesel price, solar radiation	The result indicated that the PV/diesel/battery arrangement is the optimal architecture for the sensitivities 1.1 and \$1.3/l of diesel.
[81]	Community	Rural	Surveying electrical loads and forecasting for a hybrid system in Nigeria	PV/Hydro/ Wind	Yes	–	HOMER	–	The system would have been unreliable and unstable in the eleventh year if the demand exceeded the installed capacity.
[82]	Community	Rural	Assessing a hybrid system's financial and technical possibility in a remote area.	PV/Wind/ Diesel	Yes	Battery	HOMER	Solar radiation and diesel price.	Renewable energy generation for rural communities is viable.
[83]	Community	Rural	Optimizing an autonomous hybrid system of PV/diesel with battery storage for villages	PV/Diesel	Yes	Battery	HOMER	Solar radiation, diesel price	It is an attractive solution for implementation
[84]	Residential	Rural	Analyzing of PV system and its techno-economic study for rural households in Nigeria	PV	No	Battery	HOMER/ PVSYST software	Solar radiation, diesel cost, interest rate	It was a good investment.
[85]	Community	Rural	Assessing the possibilities of solar and wind resources for hybrid systems.	PV/Wind	Yes	Battery	–	Solar radiation, wind speed	The hybrid scheme revealed that it is possible and reliable for hybrid systems.
[86]	Academic	Rural	Process optimization of college hybrid solar power system	PV	No	Battery	SPSS	–	The battery bank should be expanded.
[87]	Community	Rural	Assessing the possibility of an independent hybrid system and its techno-economic analysis for rural areas of south-eastern Nigeria	PV/Wind	Yes	Battery	HOMER	Solar radiation, wind speed, interest rate	The result revealed that the wind standalone is the cheapest in Owerri, Onitsha, and Enugu.
[88]	Community	Rural	Analyzing the abilities and economics in favor of a stand-alone hybrid system for rural north eastern Nigeria	PV/Wind	Yes	Battery	HOMER	Solar radiation, wind speed, diesel cost.	The result shows that it is competitive with grid electricity.
[89]	Community	Urban	Investigating hybrid system 5 G application in Akure Nigeria.	PV/Wind/ Diesel	Yes	Battery	HOMER	Solar radiation, wind speed, diesel cost.	The result will be economical and cost-effective.

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Table 3 (continued)

Ref	Type of load	Location	Aim	Energy sources considered	Hybrid	Energy storage considered	Method/Software	Uncertainty considered/ sensitivity analysis	Findings
[90]	Academic	Urban	Hybrid solar power system: campus model solution	PV	No	Battery	Manually	–	Load analysis was carried out for the whole campus, which was used to design energy requirements for the institution.
[91]	Community	Rural	Investigating the abilities of solar thermal technology in Nigeria	PV	No	Thermal Storage	–	–	The application will directly impact the masses, especially in rural areas, for the basic human needs.
[92]	Residential	Urban	Investigating the increment in solar penetration for houses in Nigeria	PV	No	Battery	Matlab	Cost of diesel, cost of energy, energy demand, solar radiation, interest rate, inflation rate	The result revealed savings in diesel, energy, and life cycle costs.
[93]	Academic	Urban	Developing a PV hybrid standalone for a laboratory in Lautech, Nigeria	PV	No	Battery	Manually	Temperature, potential difference	The result reveals that the system sustains power satisfactorily.
[94]	Agricultural	Rural	Examining the drying of agricultural products in Nigeria using CHP for electricity	Biomass	No	–	HYSYS	Interest rates, change in heat, cost of energy	The results revealed that the efficiency increased through heat recovery
[95]	Community	Rural	Analyzing the cogeneration system for losses after harvest and its techno-economic study	Biomass	No	–	AP Process Simulator	Interest rate, capital cost, temperature change, biogas composition	The study revealed that it has applications in storing other agricultural produce
[96]	Residential	Rural	Achievement of predicting PV generation using meteorological year dataset	PV	No	Battery	Finkelstein–Schaffer statistics and building energy simulation	Solar radiation, wind speed, temperature, relative humidity	The result revealed that TMY data can predict the PV system performance.
[97]	Community	Rural	Investigating a hydro hybrid system for rural communities in Nigeria	PV/Hydro	Yes	Battery	HOMER	–	The result revealed that adding a micro hydro to the hybrid system will increase reliability and efficiency.
[98]	Community	Rural	Examining the potentials of hydrokinetic energy in rural Nigeria	Hydro	No	Battery	–	–	The technology is feasible for rural communities.
[99]	Community	Rural	Hybrid system techno-economic and energy mix analysis for off-grid settlements	PV/Wind/Diesel	Yes	Battery	HOMER	Solar radiation, wind speed, diesel cost	According to the outcome, more than half of the world's energy production can be supplied by renewable when subjected to the three load cases.
[100]	Community	Rural	Looking at the best hybrid system for Nigeria's off-grid villages	PV/Wind/Diesel	Yes	Battery	HOMER	Interest rate, solar radiation, wind speed, diesel cost.	The result revealed that the annual real interest rate affected the optimal system arrangement.
[101]	Base Transceiver Station (BTS)	Rural	Examining PV, diesel generator and battery and their techno-economic study for base transceiver base station in Nigeria	PV/Diesel	Yes	Battery	HOMER	Diesel cost, diesel consumption, carbon dioxide emission, energy cost, net present cost renewable fraction	The study revealed that powering a BTS could be accomplished by mixing solar, diesel, and batteries.
[102]	Community	Rural	Challenges facing wind energy in Nigeria	Wind	No	–	Matlab	–	Recommended for small communities living in isolated areas
[103]	Community	Rural	PV, generator, and battery hybrid system optimization for communities in Niger Delta Nigeria	PV/Diesel	Yes	Battery	HOMER	Solar radiation, cost of fuel	The system is feasible.

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Table 3 (continued)

Ref	Type of load	Location	Aim	Energy sources considered	Hybrid	Energy storage considered	Method/Software	Uncertainty considered/sensitivity analysis	Findings
[104]	Residential	Rural	Investigating with PV electricity source and their domestic load demand	PV/Diesel	Yes	Battery	HOMER	Solar radiation, energy demand, discount rate, depreciation, PV cost	The result revealed that it is better to use PV against the existing diesel generators
[105]	Agricultural	Rural	Advancement and designing a hybrid PV electric sliced vegetable dryer	PV	No	–	Regression Equation Model	Energy consumption, change in temperature	The study revealed that the research is feasible.
[106]	Agricultural	Rural	Profitability and energy gaps of semi-mechanized and traditional rice production technologies in north central	Diesel	No	–	Sample Determination	–	According to the study, farmers should use mechanization for pre-planting tasks.
[107]	Healthcare	Rural	Examining a remote healthcare center in Nigeria and its techno-economic study	PV/Biomass/Diesel	Yes	Battery	HOMER	Load demand, cost of components and sizes, solar radiation, diesel price, interest rate	Economically sensible and for hospitals with similar loads and weather circumstances, it is technically feasible.
[108]	Healthcare	Rural	Comparing a hybrid system with solar, diesel, and batteries and a diesel generator only for rural electricity provision	PV/Diesel	Yes	Battery	HOMER	Diesel consumption, carbon dioxide emission	The result revealed that fuel consumption & carbon dioxide emissions were reduced.
[109]	Community	Urban	A decentralized PV, wind, and hybrid biogas system in northern Nigeria is being evaluated.	PV / wind/ biogas	Yes	Battery	HOMER, Microsoft Excel, and Ganzleitliche Bilanz	Reduced net present costs, leveled energy cost, component sizing, energy supply, and operating costs.	The study suggested a remedy for Nigeria's energy crisis and regular supply.
[110]	Agricultural	Rural	A DC nano-grid's modeling and techno-economic sizing for off-grid nomadic settlements in central Nigeria	PV/Wind	Yes	Battery	Particle Swarm Optimization (PSO)	Reduced energy consumption, reduced cost of electricity, reduced load demand.	A PV/battery DC Nano grid was the nomadic settlements' most viable and economical option.
[111]	Agricultural	Rural	Analyzing PV, diesel hybrid system for agricultural progress and its techno-economic study	PV/Diesel	Yes	Battery	HOMER	Solar radiation, diesel price, cost of operation, energy cost, net present cost, operating cost	The result revealed that adding renewable energy to power agricultural activities will primarily assist in food security.
[112]	Systems	–	Problems and Opportunities in the hybrid system in Sub-Saharan Africa with a focus on Nigeria	PV/Wind/Diesel	Yes	Battery	–	–	The result revealed that it solves the world's energy crisis.
[113]	Agriculture	Rural	Analyzing the possibility of a PV battery system for a farm	PV	No	Battery	HOMER	Cost of diesel, net present cost, energy cost, operation cost	The result shows that the energy cost while using the proposed energy system is lower, which will lower the energy expenditure.
[114]	Community	Rural	An assessment of renewable energy potentials in Nigeria using biogas	Biogas	No	–	Simulink	–	It is feasible and has the potential for power generation in Ikole-Ekiti.
[115]	Community	Rural	Sizing and optimization of an off-grid hybrid system for rural communities	PV/Diesel	Yes	Battery	HOMER	net present cost, load demand	The proposed system has more outstanding economic merit
[116]	Residential	Rural/Urban	Investigating energy development in Nigeria's residential sector	PV/Wind/Hydro/Diesel	Yes	Battery	TIMES/TINIREES	Cost of energy, environmental emissions	The result revealed that due to fuel costs, households should make the right choices

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Table 3 (continued)

Ref	Type of load	Location	Aim	Energy sources considered	Hybrid	Energy storage considered	Method/Software	Uncertainty considered/ sensitivity analysis	Findings
[117]	Community	Rural	Predicting solar radiation in Nigeria using adaptive neuro-fuzzy	PV	No	–	ANFIS	Solar radiation	when selecting cooking fuel. The study revealed that the reliability will be tremendous if the solar radiation is modeled using the proposed methodology.
[118]	Agricultural	Rural	Energy requirements for drying of agricultural products	PV	No	–	Matlab	–	The minimum energy equipment for drying moisture laden sliced crops was found to be between 4.22 and 24.99 MJ/kg water removed.
[119]	Agricultural	Rural	Profitability and energy gaps of semi-mechanized and traditional rice production technologies in rural areas.	Diesel	No	–	Sample Determination	Cost of production, Energy expended	Profit is higher in the semi-mechanized sector than in the traditional method.
[120]	Community	Urban	An assessment of renewable energy readiness in Nigeria	PV/Wind	Yes	–	–	–	Nigeria is ready for renewable energy implementation.
[121]	Agricultural	Rural	A techno-economic assessment of PV systems for irrigation purposes in agriculture	PV	No	Battery	HOMER	Discharge rate, Pump Power, Life cycle cost, system head.	The study revealed the variations in the system head against solar radiation.
[122]	Community	Rural	Assessment of PV and biomass sources of energy for a rural community	PV/Wind/ Diesel	Yes	Battery	HOMER	Cost, biomass output, solar radiation	The study revealed the optimum configuration as well as the level of its feasibility.
[123]	Community	Rural	An assessment of PV and hydro integration for community purpose	PV/Hydro	Yes	Hydraulic	HOMER	–	The finding revealed that the designed hybrid system could provide water for the community.
[124]	Community	Rural	An assessment of PV and wind sources of energy for the provision of power for a community	PV/Wind	Yes	–	RETScreen	Energy cost	The findings in the study revealed that the hybrid configuration is fit for the community.
[125]	Community	Rural	Optimal design of a hybrid system for a rural community	PV/Wind/ Diesel	Yes	Battery	HOMER	–	The study revealed that energy costs in Nigeria are lower than in Northern Nigeria.
[126]	Academic	Urban	Comparing the capabilities of grid and off-grid hybrid systems for university buildings.	PV/Wind	Yes	Battery	HOMER and RETScreen	Capital cost, wind speed, solar radiation, load demand, inflation, and discount rates.	The study revealed that providing energy for educational buildings is feasible using the proposed hybrid system.
[127]	Community	Rural	Sensitivity analysis of optimal hybrid system	PV/Wind	Yes	Battery	HOMER	Load, emission, diesel price, total net present cost, cost of energy, renewable penetration factor	The study revealed corresponding changes with all the variables with load demand except the energy cost, which had slight and inverse changes with daily load demand.
[128]	Academic	Urban	Assessment of hybrid system for the energy production for a university	PV/Wind	Yes	Battery	HOMER	–	The study revealed that providing energy to a university using a hybrid renewable energy system is economical.
[129]	Residential	Urban	Assessing the potential of hybrid solar renewable system in providing energy in southern Nigeria	PV	Yes	Battery	HOMER	–	The study revealed that adopting the hybrid energy generation system is more beneficial than the standalone one.

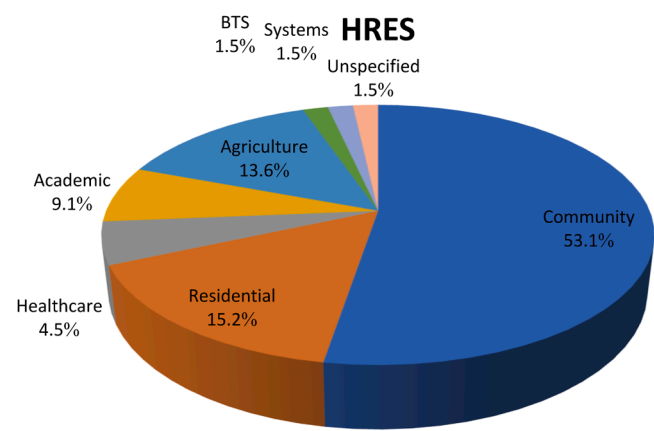


Fig. 3. Summary of various sectors that have been considered in the literature.

that these factors play a crucial role in effectively implementing these systems. The studies collectively emphasize Nigeria’s increasing preference for renewable energy, indicating a positive direction for the country’s energy future. However, they also indicate the need for further investigation into topics such as energy storage, hybridization of biomass with other renewable energy sources, and application of multicriteria decision-making methods for identifying the best energy system and legislative frameworks in future research efforts.

4. Sustainable agriculture in Nigeria

Sustainable agriculture pertains to implementing farming practices and systems that strive to fulfill present-day needs for food and textiles while safeguarding the capacity of future generations to fulfill their own needs without compromise. This strategy encompasses three primary aims, including promoting a sustainable environment, attaining economic viability, and establishing social and economic fairness. Sustainable agriculture techniques frequently encompass a range of methodologies, including but not limited to soil conservation, diminished chemical usage, integrated pest control, crop rotation, conservation tillage, and agroforestry [131]. In the Nigerian setting, sustainable agriculture tackles local issues such as unpredictable precipitation patterns, soil degradation, and the imperative of cultivating diverse crops to safeguard food sovereignty [132].

Energy plays a crucial and essential part in contemporary agriculture, influencing various facets, including the initial stages of sowing, harvesting, subsequent processing, and eventual distribution. The growing reliance on machinery, irrigation systems, and advanced agricultural technologies underscores the need for dependable energy sources. The attainment of current agricultural benefits, such as improved crop yields, improved food preservation, and efficient water utilization, would be unattainable without reliable and uncontaminated energy sources. Nevertheless, Nigeria’s lack of dependable energy in many rural agricultural areas emphasizes the urgent need to integrate energy and agriculture. Providing sustainable energy solutions is a chance to upgrade agricultural operations and uphold their environmental and economic viability.

The activities listed in Table 4 demonstrates Nigeria’s efforts to combine energy efficiency and environmentally friendly farming methods. By implementing these initiatives, Nigeria hopes to achieve its goals of increased food security, lessened environmental damage, and rural-area economic development.

5. Opportunities for powering the agricultural sector with renewable energy

The increasing requirement for energy and the concern to reduce emissions attributed to conventional energy sources has led to rising

Table 4
Selected sustainable agriculture initiatives in Nigeria.

Ref	Sustainable agriculture initiatives	Description	Locations
[133]	Solar Powered Pumping System for irrigation	Introduced to ensure consistent crop water supply, irrespective of seasonal rainfall variations. Improved yields and allowed for year-round farming.	Kaduna State
[134]	Greenhouse Farming	Establishment of controlled farming environments powered by renewable energy sources. Optimized growth conditions, reduced chemical inputs, and increased crop productivity.	Ogun State
[135]	Organic Farming Initiatives	Farmers are trained in organic farming techniques, emphasizing natural pest control, composting, and crop rotation to enhance soil health and reduce chemical use.	Various parts of Nigeria
[136]	Integrated Farming Systems	Combines crop and livestock farming to optimize resource use and improve farm productivity. Smallholder farmers integrate fish farming with rice cultivation, using fish pond water for irrigation.	Benue state
[137]	Biogas from Agro-waste	Projects focus on converting agricultural waste, especially from cattle, into biogas. Provides a clean energy source and addresses waste management challenges.	Various parts of Nigeria
[138]	Community Seed Banks	Establishment of seed banks in areas vulnerable to climate change to conserve and distribute indigenous and drought-resistant seed varieties, ensuring continued agricultural productivity.	Various locations in Nigeria

attention to other energy sources, such as renewable energy [139]. Renewable energy is in various technologies and can be used to solve a wide range of energy needs. It can provide and augment the required energy needs with very low environmental impact and greenhouse gas emissions, thereby reducing the effect of energy production on climate change [140].

In recent times, energy demand has increased tremendously in the agricultural sector [141] where energy is required to meet the energy needs of food production such as lighting, water pumping, refrigeration, drying of agricultural products, livestock rearing operations, land preparation and fertilization, irrigation, and many others. Renewable energy technologies are now being utilized to provide energy for the agricultural sector. It can solve the energy problem while mitigating the challenges associated with energy production through fossil fuels. Therefore, applying renewable energy to solve agricultural issues cannot be overemphasized as it has enormous benefits. To this end, to encourage sustainable, affordable, and clean energy as SDG 7 has promoted, improving energy production, conversion, storage, and end-use technologies is pertinent. This can be achieved through increased energy efficiency and less expensive technologies to achieve a cleaner, cheaper, and bountiful amount of energy to ease the problem of energy poverty.

Renewable energy can be used to produce clean energy from sources such as sunlight, wind, biomass, hydro, and geothermal heat. While applying the various energy conversion devices and systems [142,143]. Generating energy through renewable energy is clean, relatively inexpensive, and has minimal environmental effects when compared to fossil fuel, and its energy resources can be found everywhere [144]. Though

there are shortcomings associated with renewable energy usage, such as low intensity and intermittence of some of the renewable energy sources, for example, solar, wind, wave, and tidal energies are dependent on weather conditions and highly intermittent [143], there are ways of eliminating or minimizing this problem. This can be done through thermal storage, batteries [145] or any other storage device for energy storage for the down period and the combination of several renewable energy sources.

Renewable energy is the most reliable, efficient, and cost-effective way to solve contemporary agricultural energy needs. This is because they are well-designed, simple, and cost-effective to maintain and can produce energy that will increase agricultural productivity and economic empowerment. This technology can be implemented through solar, biomass, and wind to provide electricity for farming activities. The electricity produced can be used in heating, pumping, crop drying, lighting, and refrigeration in the farm facilities. Hybridization of the system will produce more energy consistently and remove the limitations associated with intermittency and other effects of standalone renewable energy systems. It has the advantages of no fuel consumption, low running costs, extensive long life, reliability, low maintenance, and clean energy; it reduces air and noise pollution and is free of greenhouse gas emissions. It also has the advantage of proximity to its energy resources, as in the case of biomass, solar, and wind. This is because they are all readily present on the farm for utilization as resources for energy production and not waste.

The interests of various governments, research and academic institutions, corporate bodies, and end consumers in renewable energy technology indicate that this technology will be developed for the production, conversion, storage, and conservation of energy resources from renewable energy and will continue to gain attention [146]. At present, the efficiency of some renewable energy technologies is not yet optimized. In contrast, the demand for renewable energy has continued to increase globally due to clean, environmentally friendly, futuristic, and inexhaustible opportunities [147,148].

Implementing Hybrid Renewable Energy Systems (HRES) within agricultural contexts tackles pressing energy requirements in farming and agro-processing, enhancing the sustainability, efficacy, and ecological compatibility of agricultural practices. As seen in Table 5, various proposed solutions can potentially improve both farm productivity and rural livelihoods by reducing energy costs and creating additional income-generating opportunities. Implementing these Hybrid Renewable Energy Systems (HRES) solutions demonstrates their efficacy in meeting diverse energy requirements within the agricultural sector, fostering sustainability, enhancing efficiency, and ensuring economic viability in rural farming operations.

6. Technological advancements in hybrid renewable energy systems for agricultural applications

In recent times, the advent of Hybrid Renewable Energy Systems (HRES) technology has brought about a significant shift in the agricultural landscape, presenting novel prospects for implementing sustainable, efficient, and ecologically conscious farming methods. Implementing advanced solar technologies has resulted in notable enhancements in the efficacy of photovoltaic cells and reduced associated expenses [155]. Consequently, solar energy has become more readily available and applicable to agriculture. This technological advancement guarantees that photovoltaic cells with high efficiency and solar tracking systems may optimize energy generation from sunlight [156], hence offering a dependable and uninterrupted power supply. Concurrently, implementing energy-efficient farm equipment, specifically engineered to be compatible with hybrid renewable energy systems (HRES), significantly transforms agricultural methodologies. This includes utilizing electric tractors and machinery that operate on renewable energy sources, resulting in notable reductions in fuel consumption, emissions, and operational expenses.

Table 5
Selected agricultural energy needs.

Ref	Agricultural energy needs	HRES solutions
[149]	Solar-Powered Irrigation	Utilize photovoltaic panels to generate electricity for water pumps. Ensures consistent and cost-effective irrigation for crops.
[150]	Crop Processing and Drying	Harness solar and biomass energy for crop processing equipment. Solar dryers efficiently reduce moisture content, extending shelf life—biomass energy from agricultural waste for milling and grinding.
[151]	Cold Storage and Refrigeration	Implement solar-powered refrigeration systems. - Preserve perishable agricultural products and reduce food waste.
[145]	Poultry and Livestock Farming	Solar panels power lighting and ventilation in animal housing facilities. Biomass energy for heating in brooding and livestock shelters.
[152]	Agro-Processing Industries	Hybrid energy systems (solar, wind, biomass) supply electricity to agro-processing units. - Reliable and clean energy for industrial-scale processing.
[153]	Water Pumping for Livestock and Aquaculture	Solar water pumps and wind-powered pumps provide sustainable water sources.
Reduce reliance on diesel generators or manual labor.		
[154]	Biogas for Cooking and Heating	Convert agricultural waste into biogas for cooking and heating. - Reduce indoor air pollution and energy costs.

Energy storage technologies, such as improved flow and lithium-ion batteries, have become popular because they provide a consistent energy source even when solar and wind energy output is scarce. These kinds of energy storage devices are essential to maintaining agricultural activities. This transition has made smart grids necessary since they allow for effective load management, distribution of energy, and system stability. Smart grids are crucial to the agricultural sector because they balance supply and demand for energy [157], providing a steady and dependable power source for critical operations like processing, storage, and irrigation and reducing energy waste.

Automation technologies have transformed agriculture by enabling real-time monitoring and management of numerous farming processes [158]. These technologies are driven by the Internet of Things (IoT) and artificial intelligence (AI) [159]. Increasing crop yields, cutting energy use, and optimizing resource use is how precision agriculture, self-driving tractors, and intelligent irrigation systems raise agricultural production [160]. Furthermore, precision agriculture has made precise monitoring and administration of agricultural operations possible in conjunction with IoT technology and smart sensors [161]. This has ensured effective resource allocation and crop health management. Blockchain technology has allowed consumers to track the origin and quality of products [162], bringing transparency and traceability to the agricultural supply chain.

In addition, blockchain infrastructure powered by HRES can guarantee environmental efficiency and transparency [163]. The biogas output from organic waste and collected methane emissions for energy production have improved thanks to advances in biogas improvement, especially in optimizing anaerobic digestion processes [164]. This has enhanced sustainability and reduced greenhouse gas emissions. When taken as a whole, these developments herald a paradigm change in agriculture, pointing to a time when technology and sustainable energy

will combine to increase production, lessen environmental effects, and guarantee food security.

7. Economic and environmental impacts of hybrid renewable energy systems in agriculture

Incorporating Hybrid Renewable Energy Systems (HRES) into agricultural operations has reshaped the economic landscape of agriculture and provided numerous tangible benefits [165]. First, the declining prices of renewable technologies, particularly solar panels and wind turbines, make the initial investment in HRES more affordable. HRES offers significant long-term cost savings compared to conventional energy sources such as diesel generators by reducing monthly energy expenditures and operational expenses [166,167]. In addition, renewable energy systems typically have reduced maintenance costs and longer operational lifetimes, resulting in a greater return on investment for farmers over time [168]. Integrating energy storage solutions, such as lithium-ion batteries, ensures uninterrupted power supply, thereby mitigating losses caused by power outages, particularly in post-harvest processing. In addition, numerous governments and organizations offer incentives, tax exemptions, and subsidies to encourage using renewable energy in agriculture, thereby enhancing its economic appeal. By reducing producers' reliance on fluctuating fossil fuel prices, HRES provides a more stable and predictable energy expenditure.

Integration of HRES into agriculture offers substantial environmental benefits that extend far beyond economic implications [169]. As HRES essentially eliminates the need for fossil fuels in agricultural operations, the primary benefit is a substantial reduction in greenhouse gas emissions. This reduction is essential for combating climate change and reducing the carbon footprint of agricultural practices. In addition to reducing emissions, solar-powered irrigation, wind-powered machinery, and biogas systems promote the conservation of vital resources such as water and soil. By transitioning to HRES, farms can reduce water loss through efficient irrigation systems and decrease soil degradation [170] by reducing chemical fertilizers, frequently produced through energy-intensive processes. In addition, using renewable energy in agriculture promotes biodiversity by reducing pollution and habitat devastation [171], which are typically associated with the extraction and use of nonrenewable energy. HRES guarantees the sustainability and resiliency of ecosystems and enhances the capacity of agriculture to adapt to changing climatic conditions by creating an environmentally friendly agricultural landscape [172]. Integrating Hybrid Renewable Energy Systems (HRES) with agricultural practices has significant economic and environmental implications [7]. In addition to facilitating the development of a more economically viable and environmentally sustainable agricultural industry, these initiatives provide substantial contributions towards the overarching objectives of environmental preservation and climate change mitigation. The following subsection provides a simple case for adopting renewable energy for agricultural purposes.

7.1. Case study

The techno-economic and environmental feasibility of a hybrid renewable energy system—which consists of a battery storage system and a freestanding PV, wind, biomass, and battery renewable energy system for a grid-independent farm facility, is briefly assessed in this section (Fig. 4). The Federal University of Agriculture, Abeokuta, Ogun State, Nigeria's Bill and Melinda farm facility served as the domain for the case study for this research. The agricultural facility is situated on the university property. The hatchery, cold room, incubator, incubator boxes, hatchers' room, chicks' room, brooding unit, processing unit, breeder unit, growers' unit, pasteurizing unit sections, restrooms, and administrative blocks with offices are just a few of the areas that make up the farm facility.

Renewable energy integration in rural agriculture is directly

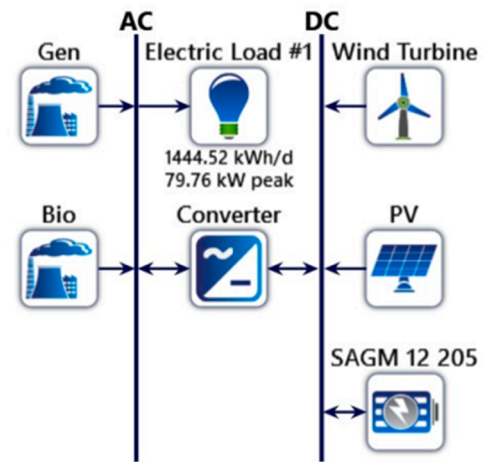


Fig. 4. Proposed system architecture.

impacted by the quality of the data collected [173]. Robust data allows for more accurate system design and optimization and improves the economic viability and overall success of sustainable energy solutions in this critical sector [174]. To implement the case study, it is essential to collect meteorology, load data, and biomass feedstock data. Meteorological data, including global sun radiation, wind speed, and temperature, was obtained from the National Space Research and Development Agency (NASRDA). Furthermore, data on biomass feedstock (plant and animal wastes) were collected during on-site visits to the farm, allowing the HOMER software to calculate electricity output from a biomass gasifier. The farm's electrical load profile was determined by analyzing the variation over time during visits to the farm site. Data on the farm's electrical equipment, including ratings and usage patterns, was obtained to calculate the farm's energy usage. This load profile data was critical to enhancing the sizing of energy sources and battery storage.

Fig. 5 illustrates the farm's load profile. The power rating of the various electrical appliances on the farm, as well as their frequency and duration of use throughout a 24-hour period—which is the farm's operating hours—were used to construct the load profile shown in Fig. 5. The farm uses 1386.85 kWh of energy every day on average, or 506,200.25 kWh per year. A 5 %-time step and daily random fluctuation were considered to increase the load demand's realism. Thus, it is calculated that the yearly energy demand is 527,250 kWh, and the daily load demand is 1444.52 kWh. The suggested energy system was simulated using HOMER. All costs in this case study are presented in United States Dollars (\$).

According to the results obtained from HOMER, the proposed system consists of a solar PV capacity of 1156 kW, six wind turbines, each with a capacity of 1 kW, a biogas generator with a capacity of 50 kW, a total of 1235 strings of lithium-ion batteries with a combined capacity of 3244 kWh, and an 85 kW converter. This architectural design incorporates a variety of sustainable energy sources to fulfill the projected annual energy demand of 526,851 kWh. The solar photovoltaic (PV) system

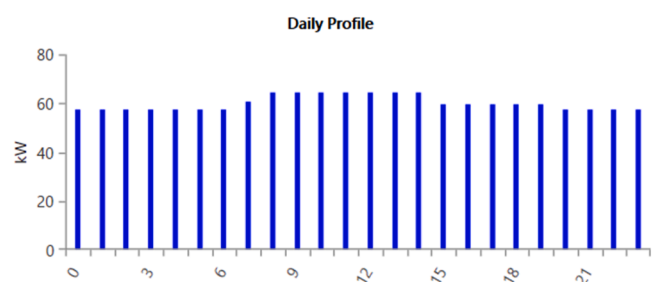


Fig. 5. Energy demand for the farm.

produces a yearly output of 1514,648 kWh, accounting for 98.3 % of the overall renewable energy generation of 1541,240 kWh per year. The biogas generator with a power output of 50 kW generates an annual energy production of 25,714 kWh, accounting for approximately 1.67 % of the whole electricity generation. The six wind turbines, each with a power output of 1 kW, collectively produce an annual energy production of 878 kWh. This amount accounts for approximately 0.06 % of the overall renewable energy generation. Overall, solar photovoltaic (PV) technology is responsible for the largest share of electricity generation, with wind power, biogas generators, and batteries playing supporting roles in addressing the challenges of unpredictability, dispatchability, and storage. The contribution of biogas generators and wind turbines to energy production is relatively minimal; nevertheless, these power sources offer additional advantages such as enhanced fuel diversity, decreased variability, and the capacity to deliver dispatchable power.

Throughout the project’s anticipated 25-year lifespan, the total net present cost is projected to be \$1782,602. The solar PV system is responsible for 42 % of total net present costs, or \$754,151. This is due to the significant initial investment required by a PV system. The cost of the battery storage system (\$942,279) accounts for the majority (53 %) of the total net present costs. Total present costs for the biogas generator come to \$34,696 (or around 2 % of the budget); the total includes a \$25,000 initial investment plus \$0 in fixed costs per year in O&M. The generator’s dispatchable power is valuable, even with the increased O&M costs, this reflects their low generation and capacity factors. When factoring in inflation, the wind turbines only add \$8897 (0.5 %) to the total net present cost. The 85-kilowatt converter has a total life-cycle cost of \$42,579.

The levelized cost of energy is estimated at \$0.26/kWh, competitive for an off-grid solution. The battery bank will also need to be replaced after 10.2 years; at this point, the cost will be roughly 60 % of the initial investment. With a present worth of \$996,161, an annual worth of \$77,057, a return on investment of 10.9 %, an internal rate of return of 15.2 %, a simple payback time of 5.84 years, and a discounted payback time of 7.22 years, the PV/wind/biomass/battery system outperformed the use of diesel generator system integrated with biomass. The figures show the financial sustainability of the investment in PV, wind, biomass, and battery renewable energy systems and the time needed to recoup its expenses. Given the positive present worth, reasonable payback period, and an attractive annual rate of return, the metrics indicate that investing in a renewable energy project is expected to be more profitable than investing in a diesel generator-biogas energy system.

The system achieves 100 % renewable penetration, which significantly reduces the carbon footprint of the farm’s electric loads. Furthermore, excess production equaling 179 % of the load can be sold to the grid, generating cash for the farm. Total CO₂ emissions are predicted to be 16 kg per year, with the biogas generator contributing 10 kg per year. Biogas combustion in a generator emits 1.64 kg/yr of CO, 0.07 kg/yr of unburned hydrocarbons (UHCs), and 0.0098 kg/yr of particulate matter (PM). These emissions can be reduced using adequate biogas cleaning and engine combustion methods. The combustion of the biogas

generator produces a relatively small amount of nitrogen oxides (NO_x) at 1.54 kg/yr. At high temperatures, NO_x is formed from nitrogen and oxygen. NO_x emissions can be reduced by using lean-burn engine designs. Biogas feedstock sourced from the farm’s organic waste streams provides excellent on-site synergy. However, monitoring feedstock availability would ensure enough year-round biogas output to keep the generator operational and performing well.

Table 6.

8. Policy, regulatory, and financing considerations

8.1. Government policies and incentives supporting hybrid renewable energy systems in agriculture

Governments across the globe are increasingly acknowledging the significance of sustainable energy solutions in the agricultural sector. They are proactively enacting rules and providing incentives to encourage the integration of hybrid renewable energy systems (HRES). These policies encompass various forms of financial incentives, such as direct subsidies or rebates for the acquisition and installation of renewable energy systems, tax exemptions or deductions for investments in home renewable energy systems (HRES), and feed-in tariffs that enable farms to sell surplus energy back to the grid at higher rates. Farmers may access research funding and technical assistance programs to facilitate their comprehension and implementation of optimal HRES solutions tailored to their specific requirements. In addition, governmental entities frequently implement awareness initiatives to provide farmers with knowledge regarding the advantages of using renewable energy sources in agricultural practices, particularly emphasizing the economic and environmental benefits. These measures aim to decrease the initial financial obstacle farmers face and enhance the economic appeal of transitioning to HRES (Renewable Energy Systems).

8.2. Challenges and opportunities in financing hybrid renewable energy systems projects for agricultural use

Although there is no denying the long-term advantages of HRES in agriculture, small-scale farmers may find it challenging to make the initial expenditure. Because of the perceived dangers involved or their lack of experience with technology, traditional financial institutions may be reluctant to lend money for renewable energy projects. With the emergence of green funding and microfinancing organizations focusing only on renewable energy projects, this situation is beginning to change. New and creative ways to finance projects, such as crowdsourcing and community-based financing methods, are also emerging. These methods combine funds to support HRES initiatives and distribute the gains to donors. Furthermore, realizing that HRES projects have the potential to spur sustainable agricultural growth, international organizations and non-governmental organizations (NGOs) are intervening to offer grants or low-interest loans for HRES projects in developing nations.

Table 6
Comparison of the present case study with similar studies in the literature.

Ref	Location	Load demand	System considered	COE	TNPC
[175]	Algeria	4.25 kW	Solar PV	0.068\$/kWh	Not stated Smottic Redcrib
[176]	India	71 kW	Diesel generator, biomass, and solar PV	\$0.032/kWh	\$76,837
[177]	India	159.2 kW	Small-scale hydropower, solar photovoltaic systems, wind turbines and bio-diesel generators	0.420/kWh	\$6,73,147
[178]	India	10 kW	Solar PV biogas, biomass, battery	INR 17.51/kWh	INR 824,067
[113]	Nigeria	0.58 kW	Solar PV, battery	\$0.535/kWh	\$17,616, and
This case study	Nigeria	79.76 kW	Solar PV, wind turbine, biomass, diesel generator and battery	\$0.26/kWh	\$1782,602

8.3. Regulatory frameworks and barriers

The regulatory environment significantly influences the adoption of HRES in agriculture. Renewable energy technologies are adopted more quickly in supportive regulatory environments with well-defined policies and efficient procedures. However, incorporating HRES can be hindered when regulatory procedures are unclear or burdensome. HRES investments by farmers may be discouraged by bureaucratic obstacles, protracted permission processes for grid connections, or ambiguous land rights. HRES solutions may also become less accessible due to import limitations or high tariffs on renewable energy technologies. Governments must develop definite, encouraging legislative frameworks that prioritize adopting renewable energy in agriculture to overcome these obstacles. Regulations that promote innovation and long-term growth in the agriculture sector can be established through cooperative efforts of policymakers, industry players, and rural communities. Although there are clear obstacles in the governmental, regulatory, and financial aspects of implementing renewable energy systems (HRES) in agriculture, several opportunities and active initiatives worldwide indicate a promising transition. Continued cooperation among many stakeholders will be crucial in guaranteeing the integration of HRES into forthcoming agricultural practices.

9. Challenges, future directions, and practical implication in the integration of hybrid renewable energy systems in agriculture

The potential for sustainable and efficient farming methods is enormous when Hybrid Renewable Energy Systems (HRES) are integrated into agriculture. However, several obstacles must be overcome to make this transformation successful and maximize the benefits for farmers and the environment. This section discusses the challenges and future directions in integrating HRES in agriculture.

9.1. Challenges

1. **Economic Concerns:** The enormous investments involved in acquiring and installing Hybrid Renewable Energy Systems (HRES) facilities is a major reason why farmers, especially farmers on a smaller scale, do not adopt methods of electricity production. Significant investments in advance are required for renewable energy means of energy generation, such as solar and wind. Also, constraints to flexible financial access opportunities make the situation more complex. Many financial organizations have yet to come to terms with the enormous opportunities associated with HRES. This lack of understanding on the part of the financial organizations has created problems for farmers while trying to secure funds for investments in HRES because the financial organizations view investment in the area as risky.

2. **Regulatory Hurdles:** Using renewable energy for agricultural purposes is often hit with complex regulations. Complex and contradictory regulations and processes can frustrate the adoption of HRES in the agricultural sector. One of these complex regulations includes ambiguous land ownership and rights. This can prevent farmers from the installation of HRES facilities on their farms. Also, restricted access to renewable energy technologies due to high import duties and taxes will likely hinder access to the sector's most recent and sophisticated technologies.

3. **Technical Barriers:** Many farmers do not have the technical competence to choose the best renewable energy systems for subsequent installation and maintenance. This gap in manpower deficiency often leads to the design of a non-optimal system for use, decreasing the system's efficiency. The absence of the required knowledge and skills to make decisive choices will make it difficult for farmers to align with the most suited renewable energy technology for their respective agricultural purposes. This will affect the cost of energy and the efficiency of the system.

4. **Reliability Concerns:** Reliability difficulties are associated with the

periodic availability of renewable energy resources such as solar and wind. Some farming operations, such as those requiring continuous energy for irrigation or refrigeration, cannot tolerate interruptions in power supply. Varying renewable energy resources can result in a situation whereby it will be challenging to forecast the energy demand required for agricultural purposes. Ensuring a consistent and reliable energy supply is essential for the success of HRES integration in agriculture.

5. **Selection of optimal system:** Using multicriteria decision-making (MCDM) analysis is a significant gap that provides a promising avenue for future study in integrating Hybrid Renewable Energy Systems (HRES) in agriculture. By considering many criteria and objectives, MCDM approaches can aid in resolving complicated decision-making processes.

9.2. Directions for the future

The review has made available critical introspection that emphasizes the various opportunities available for research purposes in the future in hybrid renewable energy systems (HRES) in agriculture. These directions can potentially enhance the transfer of knowledge, tackle current obstacles, and contribute to the sustainable evolution of agricultural methodologies. Future research can focus on technical, economic, social, policy, and environmental aspects to provide data-driven and actionable insights that can accelerate the transition towards sustainable agriculture powered by renewable energy. The following are potential avenues for further research:

- **Analysis for Economic Feasibility:** it would be interesting to perform a comprehensive economic feasibility assessment that will consider the various types of operations in agriculture, the sizes of farms, and the differences associated with the various regions. A systematic examination of the return on investments and financial advantage of adopting the HRES for agricultural purposes while considering other determinants such as types of crops, energy demand level, and localized energy markets.
- **Advancements in Technology:** Inventive research and developments in HRES that will address the energy demands in agriculture are also suggested. This could involve improving renewable energy system efficiency and cost for farms and agricultural purposes.
- **Solutions to Energy Storage:** Examining different advancements in energy storage systems solutions is another area of research that should take priority. This encompasses the various generations of storage systems such as flywheel systems, thermal energy storage, and batteries to address the periodic availability of renewable energy resources for renewable energy generation. This endeavor should examine the various optimal storage technologies that could be adopted for agricultural purposes.
- **Models Optimization:** The development of models for optimization and decision-making tools that will aid farmers in effectively designing and managing hybrid renewable energy systems is also suggested. The models will factor in energy needs, availability of renewable energy resources, and costs.
- **Assessments of Regional Resources:** A study to perform a comprehensive evaluation of the abilities of renewable energy resources for different agricultural regions is also essential. The study would examine the availability of renewable energy resources such as solar, wind, and biomass to direct farmers in selecting the most feasible hybrid renewable energy technologies.
- **Analysis of Regulations and Policies:** A breakdown of the effect of regulations and policies on using hybrid renewable energy systems for agricultural purposes. Subject to analysis policy frameworks that are fruitful and identify the best way to advocate integrating renewable energy into agriculture.

- **Demonstrations and Case Studies:** An analysis of comprehensive demonstrations and case studies of various projects that can be adopted using a hybrid renewable energy system in real-world agricultural scenarios is also essential. A deep understanding and an emphasis on success stories would help farmers in the proliferation of the technology.
- **Community-Based Initiatives:** Future studies are expected to examine models that can help farmers invest and take advantage of hybrid renewable energy systems through cooperative initiatives. Encouraging collaborative efforts to implement renewable energy is a remarkable strategy for achieving sustainable development. These projects can manifest in several ways, requiring cooperation among individuals, communities, businesses, and government entities.
- **Environmental Impact Assessment:** Investigating the effects of integrating the hybrid renewable energy system into agriculture on the environment is also an area for future studies. This investigation includes the reduction of greenhouse gas emissions, biodiversity improvement, and soil conservation—an assessment of the long-term effect of sustaining hybrid renewable energy system usage.
- **Integration for Agricultural Precision:** An examination to determine the combining effects of hybrid renewable energy systems and agricultural precision technologies can also be considered. These technologies include Internet of Things (IoT) sensors and artificial intelligence (AI). They are used to motivate farm management. Examine ways by which integration will maximize the use of resources to improve agricultural productivity.
- **Measurement of Energy Efficiency:** An investigation to determine the effect of agricultural actions and technologies that are energy efficient, which can cause a decrease in the farm's energy consumption. Examine the privileges associated with conserving energy while adopting renewable energy means of energy generation.
- **Social Effects and Rural Development:** A study to determine comprehensively the socioeconomic effect of using hybrid renewable energy systems for agricultural purposes in rural communities. Future studies can investigate ways by which renewable energy can positively impact the creation of jobs, eradicate poverty, and improve the standard of living.
- **Comparative Studies:** An investigation to evaluate comparative studies among various renewable energy systems technologies to ascertain the most feasible renewable energy combination for different agricultural load scenarios is essential. The study is expected to compare the merits and demerits of solar, wind, biomass, and other sources.
- **Multi-Criteria Decision Making:** Further research should investigate utilizing multicriteria decision-making (MCDM) analysis to optimize decision-making procedures for integrating Hybrid Renewable Energy Systems (HRES) within the agricultural sector. Multi-criteria decision-making (MCDM) can provide valuable support in various areas, such as technology selection, site evaluation, resource allocation, risk assessment, regulatory compliance, and environmental impact assessment. By employing MCDM, farmers can make well-informed decisions regarding adopting sustainable and environmentally friendly Hybrid Renewable Energy Systems (HRES).

9.3. Practical implications of the study

Hybrid Renewable Energy Systems (HRES) have the potential to change agricultural practices, especially in rural regions drastically. This highlights the urgent need for further investment and governmental assistance in the HRES industry. Governments should put Incentives and finance options in place to hasten the general adoption of HRES. Farmers who transition to HRES can experience significant increases in income and production while lessening their environmental effects. A critical factor in encouraging wider deployment is disseminating practical case studies and demonstrations. Allocating resources towards Research and

Development (R&D) to enhance Hybrid Renewable Energy Systems (HRES) technologies for agricultural applications, such as optimizing the performance of solar panels and wind turbines, can enhance viability and gain endorsement from farmers. Programs centered around rural development should integrate the provision of renewable energy as a crucial catalyst for promoting sustainable agricultural practices and fostering economic empowerment. Providing training programs for farmers that cover the sizing, selection, operation, and maintenance of hybrid renewable energy systems (HRES) can efficiently overcome technical obstacles and maximize their usage.

Reliability is improved, and intermittent problems are addressed by strategically combining several renewable resources, such as solar, wind, and biogas, adapted to local conditions. Applications of HRES that are useful and scalable include water pumping, cold storage, crop processing machinery, and solar-powered irrigation. Advances in block-chain technology have the potential to bring traceability and transparency to agricultural supply networks that run on renewable energy. In summary, there is a great deal of promise for rural development, income growth, carbon footprint reduction, and environmental sustainability when HRES is combined with agriculture.

10. Conclusions and prospects

Hybrid renewable energy systems (HRES) offer a possible path toward environmentally responsible and productive farming. This study examined the benefits and drawbacks of using HRES in agriculture, focusing on the necessity of strong policy backing, financing mechanisms, and regulatory frameworks. The research also showcases the substantial capacity of hybrid renewable energy systems (HRES) to revolutionize agricultural practices by providing environmentally friendly, dependable, and economically efficient power. Nevertheless, it is essential to acknowledge certain constraints of this review. The emphasis was predominantly on theoretical modeling rather than practical real-world case studies. Additional empirical evaluations of operational hybrid renewable energy systems (HRES) in agriculture projects would enhance the analysis.

Furthermore, the study was confined to the specific circumstances of Nigeria, thereby constraining the potential to make comparisons with other developing nations. A comprehensive assessment of the adverse health and environmental effects was not conducted, indicating additional life cycle evaluations were needed. Furthermore, the long-term performance of the system may deviate from short-term predictions. Further examination is required to thoroughly explore the aspects of policy, regulatory, and institutional elements. Also, multicriteria decision-making (MCDM) analysis emerges as an essential area for further study, as it provides a systematic method for tackling the complexities of technology choice, resource management, and environmental concerns inherent in HRES deployment. Renewable energy sources are increasingly being adopted in the agricultural sector because of their positive effects on economic viability, environmental protection, and climate change mitigation. Sustainable food production, rural development, and a greener planet are all possible with the help of agriculture powered by HRES in the future, which can be achieved with concerted efforts from governments, financial institutions, and the research community.

CRedit authorship contribution statement

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Declaration of competing interest

None.

Data availability

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